Secular coefficients mini-course
Talk 1

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1. Talk 1: Riemann zeta function and analogies
2. Talk 2: Random multiplicative functions: basic results
3. Talk 3: Random multiplicative functions: sum of the Steinhaus function
4. Talk 4: Combinatorial perspective on moments of zeta

Definition

The Riemann zeta function is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = 1 + 2^{-s} + 3^{-s} + 4^{-s} + \ldots$$

for $s \in \mathbb{C}$ with $\Re s > 1$.

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for $s \in \mathbb{C}$ with $\Re s > 1$. We shall review its properties (without proof).

Euler product

For $\Re s > 1$,

$$\zeta(s) = \prod_p \left(1 + p^{-s} + p^{-2s} + \dots\right) = \prod_p (1 - p^{-s})^{-1}.$$

Analytic continuation

ζ has a meromorphic continuation to $\mathbb{C}$ with a simple pole at $s = 1$ and residue 1:

$$\zeta(s) - \frac{1}{s-1}$$

is entire.

Functional equation

Define

$$\chi(s) := \Gamma(s/2)\pi^{-s/2}s(s-1)/2.$$

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$$\xi(s) := \zeta(s)\chi(s).$$

Then $\xi(s)$ is entire and

$$\xi(s) = \xi(1-s).$$

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Note that $\Gamma(s/2)$ has simple poles at $s = 0, -2, -4, \ldots$.
The poles cancel with zeros of $\zeta(s)$ at $s = -2, -4, \ldots$ and the factor s.

Product over zeros

We have

$$\xi(s) = \xi(0) \prod_{\rho:\, \zeta(\rho)=0,\, \rho\neq -2,-4,\ldots} \left(1 - \frac{s}{\rho}\right).$$

for an explicit real A.

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Number of zeros

$$N(T) = |\{\rho : \zeta(\rho) = 0,\, 0 < \Im\rho \leq T\}| \sim \frac{T \log T}{2\pi}.$$

Connection with primes

The connection arises from relating primes (appearing in the Euler product) to the zeros (appearing in the Hadamard product).

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Consider the logarithmic derivative of $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$:

$$-\frac{\zeta'}{\zeta}(s) = \sum_p \log p \frac{p^{-s}}{1 - p^{-s}} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

where Λ is the *von Mangoldt function*,

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k, \\ 0 & \text{otherwise.} \end{cases}$$

Connection with primes (cont.)

The series $-\frac{\zeta'}{\zeta}(s)$ converges for $\Re s > 1$, and has a meromorphic continuation to $\mathbb{C}$, with pole at every zero or pole of ζ.

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Just as Cauchy's formula expresses a coefficient of a power series as a complex integral, *Perron's formula* implies the famous "Explicit Formula":

$$\sum_{n \leq x} \Lambda(n) = \frac{1}{2\pi i} \int_{(2)} -\frac{\zeta'}{\zeta}(s) \frac{x^s}{s} ds = x - \sum_{\rho:\, \zeta(\rho)=0} \frac{x^\rho}{\rho}.$$

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Conversely, this estimate implies the Riemann hypothesis.

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Conversely, this estimate implies the Riemann hypothesis. In prime language,

$$\sum_{p \leq x} 1 = \int_2^x \frac{dt}{\log t} + O(\sqrt{x}\log x).$$

Probabilistic perspective on the Riemann zeta function

The Steinhaus function

Define $\alpha\colon \mathbb{N} \to \mathbb{C}$ to be a completely multiplicative function:

$$\alpha(nm) = \alpha(n)\alpha(m),$$

where $(\alpha(p))_p$ are iid random variables uniformly distributed on the unit circle.

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Denote

$$F(s) = \sum_{n=1}^{\infty} \frac{\alpha(n)}{n^s} = \prod_p \left(1 - \frac{\alpha(p)}{p^s}\right)^{-1}$$

for $\Re s > 1$.

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for $\Re s > 1$.
α is called the *Steinhaus random multiplicative function*.

Theorem 1 (Bohr–Jessen, 1930s)

Fix $s \in \mathbb{C}$ with $\Re s > 1/2$. Let U_T be a random variable uniformly distributed on $[0, T]$. Then

$$\zeta(s + iU_T) \xrightarrow{d} F(s)$$

as $T \to \infty$.

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More generally, given $s_1, \ldots, s_k$ with real parts $> 1/2$,

$$(\zeta(s_1 + iU_T), \ldots, \zeta(s_k + iU_T)) \xrightarrow{d} (F(s_1), \ldots, F(s_k)).$$

Intuition

For any fixed y,

$$(p^{iU_T})_{p\leq y} \xrightarrow{d} (\alpha(p))_{p\leq y}.$$

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RHS is 1 if $a_p = 0$ holds for all p, and 0 otherwise.
LHS is 1 if $a_p = 0$ holds for all p, and tends to 0 otherwise.

Haar measure

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Let $U(N)$ be the group of N-by-N complex unitary matrices U with $U\overline{U^T} = I_N$. There is a unique probability measure μ on $U(N)$ with the property: if U_N has law μ, then

$$\mathbb{P}(U_N \in X) = \mathbb{P}(AU_N \in X) = \mathbb{P}(U_N A \in X)$$

for every matrix $A \in U(N)$ ("translation-invariance"). It is called the *Haar measure*.

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No Euler product.

Theorem 2 (Bohr–Jessen for matrices (90s))

Fix u with $|u| < 1$. Let $(N_i)_{i \geq 1}$ be iid standard complex Gaussians. Then

$$\det(I - uU_N) \xrightarrow{d} F(u)$$

as $N \to \infty$ where

$$F(u) = \exp\left(- \sum_{i=1}^{\infty} \frac{N_i}{\sqrt{i}} u^i \right).$$

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More generally, given $u_1, \ldots, u_k$ inside the unit circle,

$$(\det(I - u_1 U), \ldots, \det(I - u_k U)) \xrightarrow{d} (F(u_1), \ldots, F(u_k)).$$

Intuition

Taking logarithms,

$$\log\det(I - uU_N) = -\sum_{i=1}^{\infty} \frac{\mathrm{Tr}(U_N^i)}{i} u^i.$$

By a result of Diaconis–Shahshahani:

$$\left(\frac{\mathrm{Tr}(U_N^i)}{\sqrt{i}}\right)_{i\leq k} \xrightarrow{d} (N_i)_{i\leq k}.$$

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Given a Dirichlet character, its Dirichlet L-function is

$$L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p (1 - \chi(p)p^{-s})^{-1},$$

defined for $\Re s > 1$.

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Functional equation relates $L(s, \chi)$ to $L(1 - s, \overline{\chi})$. This is similar to how $\det(I - uU)$ relates to $\det(I - u^{-1}\overline{U})$.

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Again, one can connect primes to zeros of $L(s, \chi)$ through Perron's formula, obtaining another Explicit Formula:

$$\sum_{n \leq x} \Lambda(n)\chi(n) = \frac{1}{2\pi i} \int_{(2)} -\frac{L'}{L}(s, \chi) \frac{x^s}{s} ds = -\sum_{\rho:\, L(\rho,\chi)=0} \frac{x^\rho}{\rho}.$$

One has a Bohr–Jessen theorem for L-functions: let χ_m be a Dirichlet character chosen uniformly at random from the group of Dirichlet characters modulo m. Then, for $\Re s > 1/2$,

$$L(s, \chi_m) \xrightarrow{d} \prod_p (1 - \alpha(p)/p^s)^{-1}$$

as m tends to infinity along primes.

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This leads naturally to the consideration of the zeta function of $\mathbb{F}_q[T]$:

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Functional equation: if we multiply by $(1 - q^{-s})^{-1}$ ("prime at infinity"), then there is symmetry $s \leftrightarrow 1 - s$ (up to a simple entire factor).

A much richer object is the Dirichlet L-function of a Dirichlet character $\chi\colon \mathbb{F}_q[T] \to \mathbb{C}$,

$$L(s,\chi) = \sum_{f \text{ monic}} \frac{\chi(f)}{|f|^s}.$$

Here a Dirichlet character modulo $M \in \mathbb{F}_q[T]$ is a completely multiplicative function with $\chi(f+M) = \chi(f)$ for every $f \in \mathbb{F}_q[T]$. It can be constructed by lifting a character of $(\mathbb{F}_q[T]/M)^\times$.

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Notation: $L(s,\chi) = \mathcal{L}(q^{-s},\chi)$ where

$$\mathcal{L}(u,\chi) = \sum_{f \text{ monic}} \chi(f) u^{\deg f}$$

is a *power series in u.*

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Theorem (Weil): The roots of $\mathcal{L}(u, \chi)$ lie on $|u| = 1/\sqrt{q}$, i.e., the roots of $L(s, \chi)$ lie on $\Re s = 1/2$.

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Theorem (Weil): The roots of $\mathcal{L}(u, \chi)$ lie on $|u| = 1/\sqrt{q}$, i.e., the roots of $L(s, \chi)$ lie on $\Re s = 1/2$.

Implication: $\mathcal{L}(u/\sqrt{q}, \chi)$ shares many properties with $\det(I - uU)$.

Given a nontrivial Dirichlet character χ modulo M, we may write

$$\mathcal{L}(u/\sqrt{q}, \chi) = \det(I - u\Theta_\chi)$$

for some conjugacy class $\Theta_\chi \in U(\deg M - 1)$ (according to Weil).

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Theorem 3 (Katz, 2013)

Fix the degree of M. As $q \to \infty$, we have

$$\lim_{q\to\infty} \frac{1}{\phi(M)} \sum_{\chi \bmod M} f(\Theta_\chi) = \mathbb{E} f(U_{\deg M-1})$$

for any conjugation-invariant f.

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$$\log \mathcal{L}(u/\sqrt{q},\chi) = \sum_{n=1}^{\infty} A_n(\chi)\frac{u^n}{n}$$

where

$$A_n(\chi) = \frac{1}{q^{n/2}} \sum_{\deg f = n,\, f \text{ monic}} \Lambda(f)\chi(f).$$

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On the other hand, from $\log\det(I - uU) = -\sum_{n=1}^{\infty} \mathrm{Tr}(U^n)\frac{u^n}{n}$ we have

$$A_n(\chi) = -\mathrm{Tr}(\Theta_\chi^n).$$

The equality

$$\frac{1}{q^{n/2}} \sum_{\deg f = n,\, f \text{ monic}} \Lambda(f)\chi(f) = -\mathrm{Tr}(\Theta_\chi^n)$$

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For U_N we have $\log\det(I - uU_N) = -\sum_{n=1}^{\infty} \mathrm{Tr}(U_N^n)\frac{u^n}{n}$ and Diaconis–Shahshahani proved: $\mathrm{Tr}(U_N^n)/\sqrt{n}$ tends to standard complex Gaussian as $N \to \infty$.

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In fact, $A_n(\chi_M)/\sqrt{n}$ (for χ_M random character modulo M) also tends to standard complex Gaussian by taking an appropriate limit (see exercises).

One of the main questions concerning the Riemann zeta function is the growth of its moments:

$$\frac{1}{T}\int_0^T \left|\zeta\left(\frac{1}{2}+it\right)\right|^{2k} dt = \mathbb{E}\left|\zeta\left(\frac{1}{2}+iU_T\right)\right|^{2k}.$$

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Hardy–Littlewood (1918): For $k=1$, grows like $\log T$.

Ingham (1926): For $k=2$, grows like $\frac{1}{12}\zeta(2)^{-1}(\log T)^4$.

No asymptotic results for $k \geq 3$, but number theorists expected $\sim a_k g_k (\log T)^{k^2}$ for a well-understood arithmetic constant a_k, and an elusive geometric constant g_k (e.g. $g_2 = 1/12$ and $a_2 = \zeta(2)^{-1}$).

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Theorem 4 (Keating–Snaith)

We have

$$\mathbb{E}|\det(I - U_N)|^{2k} = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+2k)}{(\Gamma(j+k))^2} \sim N^{k^2} \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}.$$

Conjecture (Keating–Snaith)

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Conjecture (Keating–Snaith)

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$$\frac{1}{T}\int_0^T \left|\zeta\left(\frac{1}{2}+it\right)\right|^{2k} dt \sim a_k g_k (\log T)^{k^2}$$

for $g_k = \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}$. *(*$g_1 = 1$*,* $g_2 = 1/12$*,* $g_3 = 42/9!$*.)*

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Heuristic for $(\log T)^{k^2}$ growth:

$$\log|\zeta(\tfrac{1}{2}+iU_T)| \approx -\sum_{p\leq T} \log|1-p^{-\frac{1}{2}-iU_T}| \approx \sum_{p\leq T} p^{-\frac{1}{2}} \cos(U_T \log p).$$

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If this were a sum of independent random variables, the variance would be $\frac{1}{2}\sum_{p\leq T} p^{-1} \sim (\log\log T)/2$. Then the central limit theorem would tell us that

$$\frac{\log|\zeta(1/2+iU_T)|}{\sqrt{(\log\log T)/2}} \xrightarrow{d} N(0,1).$$

The last limit is actually an unconditional result of Selberg (1940s). It has a matrix analogue:

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Moments of ζ should grow like

$$\mathbb{E}|\zeta(1/2+iU_T)|^{2k} \approx \mathbb{E}e^{2k\sqrt{(\log\log T)/2}Z} = (\log T)^{k^2}$$

where $Z \sim N(0,1)$. This is only a heuristic, as it is missing both a_k and g_k.

Correlation functions

k-point correlation

Given a point process $X = (X_i)_i$ (a collection of random real points), one defines its *k-point correlation function* $\rho_k^X : \mathbb{R}^k \to \mathbb{R}$

$$\mathbb{E} \sum_{i_1,\ldots,i_k \text{ distinct}} f(X_{i_1}, \ldots, X_{i_k}) = \int_{\mathbb{R}^k} f(\vec{t})\rho_k^X(\vec{t})dt_1 \cdots dt_k.$$

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Given a point process X, we say that a sequence of point processes $(X_N)_N$ tends in correlation to X if $\lim_{N\to\infty} \rho_k^{X_N} = \rho_k^X$ for every k.

Convergence in correlation often implies convergence in distribution: $\sum_i f(X_{N,i}) \xrightarrow{d} \sum_i f(X_i)$.

Theorem 5 (Dyson, 1960s)

Define a point process X_N by taking the eigenangles $\theta_i \in [-\pi, \pi)$ of a random unitary matrix U_N.
The point process $\frac{N}{2\pi} \cdot X_N$ tends in correlation and in distribution to the sine point process Sine, defined by $\rho_k(t_1, \ldots, t_k) = \det\left(S(t_i - t_j)\right)_{1 \le i,j \le k}$ where

$$S(t) = \frac{\sin(\pi t)}{\pi t}.$$

Conjecture (GUE Conjecture, Montgomery, 1970s)

Assume RH and list the zeros of zeta as $1/2 + i\gamma_n$, $n \in \mathbb{Z}$.
Consider the point process $X_T = \left(\frac{\log T}{2\pi}(\gamma_n - U_T)\right)_{n\in\mathbb{Z}}$ *where* U_T *is uniformly distributed on* $[0, T]$.
Then X_T *tends in distribution and in correlation to* Sine.

Montgomery's conjecture was originally stated as a "pair correlation conjecture", concerning ρ_2 only:

$$\mathbb{E} \sum_{i_1 \neq i_2} f((X_T)_{i_1}, (X_T)_{i_2}) \to \int_{\mathbb{R}^2} f(t_1, t_2)(1 - S(t_1 - t_2)^2) dt_1\, dt_2$$

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Consequence for primes:

$$\frac{1}{X}\int_0^X\left(\sum_{x\leq n<x+H}\Lambda(n)-H\right)^2 dx \sim H\log\frac{X}{H}$$

as $X\to\infty$.

Rudnick–Sarnak (1990s): Under RH, computed limit $\mathbb{E}\sum_{i_1,\cdots,i_k \text{ distinct}} f(X_{i_1},\cdots,X_{i_k})$ for “band-limited functions”, giving further evidence for the GUE conjecture.

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The fact that convergence in correlation and in distribution are one and the same in the GUE conjecture was recently established (Arias de Reyna–Rodgers).

Thank you for listening.